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A Bipolar Voltage-Controlled Tunable Filter

KIYOSHI FUKAHORI

Abstract—A complete monolithic state variable filter is described which has been fabricated with bipolar technology. Two-quadrant multipliers are used in a novel fashion to achieve integration time constant large enough for audio purposes. Voltage control of frequency response from 20 Hz to above 20 kHz has been achieved without sacrifice of accuracy (notch depth >50 dB, low-pass and bandpass responses accurate to better than 0.1 dB), dynamic range (>90 dB), power supply rejection (PSRR > 40 dB) or frequency drift (<200 ppm/°C). The design has been shown to be generally useful in the design of self-contained bipolar filters.

I. INTRODUCTION

HISTORICALLY precision filters have been realized using passive inductor/capacitor (*LC*) networks, despite the physical and electrical limitations of available inductors. The current ubiquity of low cost monolithic op amps has, however, brought about a gradual change to *RC* active filters. In both cases, though, frequency response can only be adjusted by altering the absolute values of the respective component values; the procedure is very complex for high-order filters.

The MOS switched-capacitor filtering technique [1], [2] has emerged in the last four years that has alleviated this

problem and monolithic realization of the filters has been made possible. However, such MOS filters require a clock input, generally tend to be noisy, and suffer from poor PSRR. In addition, they tend to suffer from the clock noise that may feed through to the filter output.

This paper describes an alternate filtering technique that makes possible the monolithic realization of filters in bipolar technology [3]. More specifically, it allows the construction of high-performance stand-alone filters that are continuously tunable; exhibit accurate frequency response; require no external precision components or clock input; have a large dynamic range, excellent PSRR, and low temperature drift. Furthermore, its absolute bandwidth can be trimmed at wafer sort to a desired value.

It was decided to use a state variable approach because of the low sensitivity to component variations, but one of the major difficulties imposed by monolithic implementation—that of obtaining suitably large time constants—had to be overcome. This time constant must be large enough to do filtering in the low audio frequency range (<100 Hz) which is not a trivial task considering the fact that the total amount of resistance and capacitance that can be integrated on a chip is extremely limited. This is the main reason that *RC* active filters usually require large off-chip passive components. The crux of this

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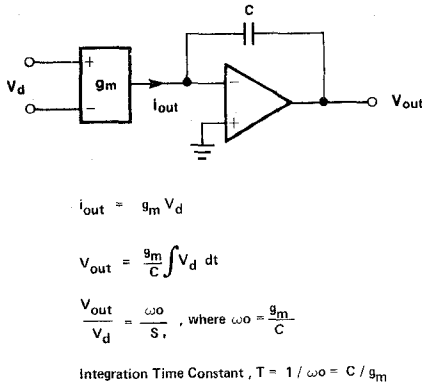


Fig. 1. Block diagram of differential integrator.

paper is to present a circuit technique to resolve this problem while maintaining the attractive features of RC active filters.

II. INTEGRATOR DESIGN

A conceptual realization of a differential integrator is shown in Fig. 1. It is nothing more than a simple transconductance amplifier followed by an integrating capacitor placed across an inverting amplifier. The current output I_{out} of the transconductance amplifier (with its transconductance equal to g_m) is given by

$$I_{out} = g_m \cdot V_d$$

where V_d is the differential voltage applied across the input terminals. The integrator output V_{out} is thus given by

$$V_{out} = \frac{g_m}{C} \int V_d \cdot dt$$

where C is the value of the integrating capacitor. In terms of the Laplace variable s ,

$$\frac{V_{out}}{V_d} = \frac{\omega_o}{s}, \quad \text{where } \omega_o = \frac{g_m}{C} \equiv \text{unity gain frequency.}$$

The integration time constant T is given by the inverse of the unity gain frequency ω_o . That is,

$$T = \frac{1}{\omega_o} = \frac{C}{g_m}. \quad (1)$$

The transconductance amplifier can be realized as shown in Fig. 2. Amplifiers U_1 and U_2 , in conjunction with Q_2 and Q_3 , respectively, form unity gain buffers and are used to transfer the differential voltage V_d across the resistor R through which the signal current i_s flows. The signal current (V_d/R) is added to and subtracted from I_o (the magnitude of the two dc current sources) at the collectors of Q_2 and Q_3 yielding collector currents of $(I_o + i_s)$ and $(I_o - i_s)$, respectively. When they are fed to the p-n-p current mirror Q_9 and Q_{10} the current output I_{out} is equal to

$$I_{out} = \frac{2}{R} \cdot V_d.$$

That is, the transconductance g_m is given by

$$g_m = \frac{2}{R}.$$

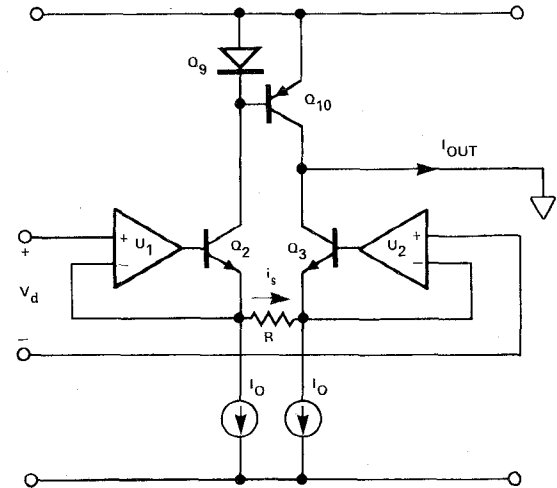


Fig. 2. Simple realization of transconductance amplifier.

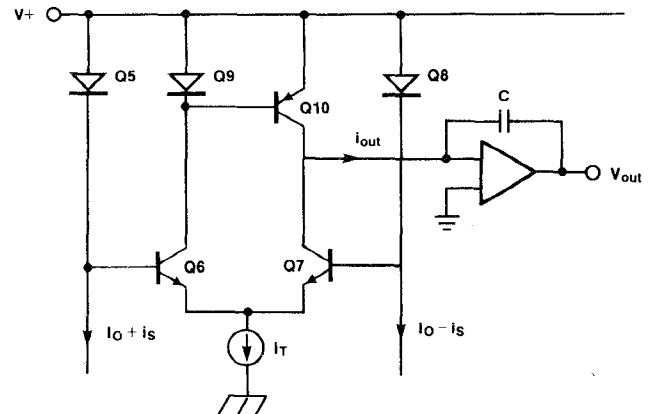


Fig. 3. Two-quadrant multiplier used to reduce transconductance or to increase integration time constant.

This in conjunction with the integrating capacitor (C) will produce an integration time constant of $RC/2$. With a practically integrable amount of R and C the magnitude of the time constant realizable with this design is rather small. For example, if

$$R = 40 \text{ k}\Omega$$

$$C = 400 \text{ pF},$$

then

$$T = 8 \mu\text{s}.$$

This is certainly not large enough to do filtering in sub 100 Hz region. Another drawback of this design is that it does not lend itself to convenient tuning.

III. NEW INTEGRATOR DESIGN

The above problems are eliminated by the use of a two-quadrant multiplier [4] as shown in Fig. 3. The collector currents of Q_2 and Q_3 in Fig. 2 are first fed to the two quadrant multiplier consisting of Q_5 - Q_8 and its outputs are then fed to the p-n-p current mirror Q_9 , Q_{10} . The effective transconductance g_{meff} is now reduced from $2/R$ by a factor equal to $I_T/2I_o$, where I_T is the magnitude of the tail current of the Q_6 - Q_7 differential pair. That is,

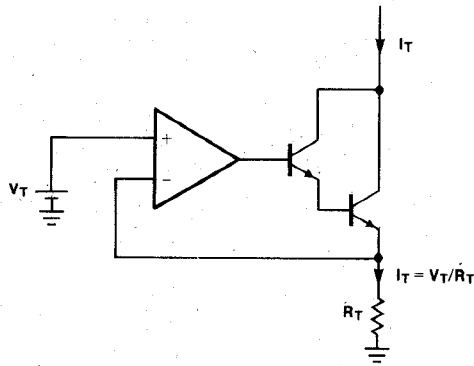
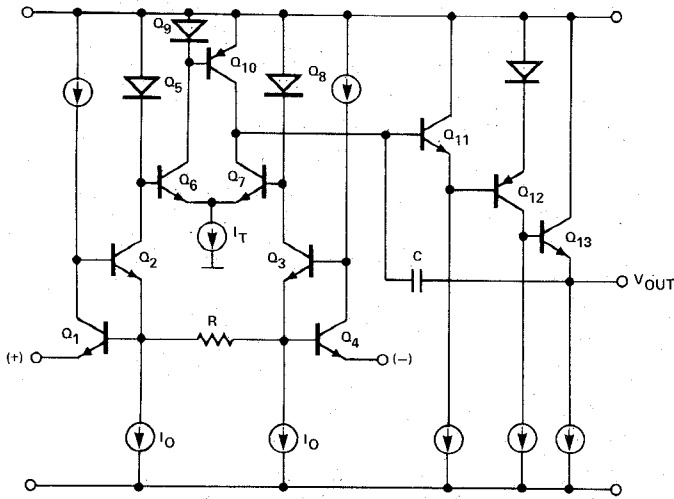
Fig. 4. V_T/I_T converter.

Fig. 5. Complete integrator.

$$g_{meff} = \frac{1}{R} \frac{I_T}{I_O} \quad (2)$$

The integration time constant T and the unity gain frequency ω_o are changed by the same factor and given as

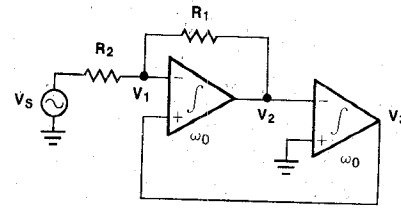
$$T = \frac{C}{g_{meff}} = RC \frac{I_O}{I_T} \quad (3)$$

$$\omega_o = \frac{1}{T} = \frac{1}{RC} \frac{I_T}{I_O} \quad (4)$$

With the same R and C values used earlier, 16 ms of time constant is obtained when the ratio of the two dc currents I_O and I_T is made equal to 1000. The integration time constant can be varied to any value by changing this ratio.

I_O and I_T are generated from two external dc voltage sources through a pair of voltage to current converters, shown in Fig. 4 for the case of I_T . The external dc voltage V_T is converted to I_T through an on-chip resistor R_T . A Darlington pair connection is used to minimize the base current loss. I_O is similarly developed from another external dc voltage V_O through another on-chip resistor R_O .

Fig. 5 shows the complete integrator. The two unity gain buffers in Fig. 2 are replaced by Q_1 - Q_2 and Q_3 - Q_4 [5]. Q_{12} is a common emitter p-n-p amplifier that provides inverting gain for the integrating capacitor. In terms of the two external dc voltages V_O and V_T , (3) and (4) can be rewritten as



$$\text{Notch : } \frac{V_1}{V_S} = \frac{(1 - 1/Q) (S^2 + \omega_o^2)}{S^2 + 1/Q \omega_o S + \omega_o^2}$$

$$\text{Band Pass : } \frac{V_2}{V_S} = \frac{-(1 - 1/Q) \omega_o S}{S^2 + 1/Q \omega_o S + \omega_o^2}$$

$$\text{Low Pass : } \frac{V_3}{V_S} = \frac{(1 - 1/Q) \omega_o^2}{S^2 + 1/Q \omega_o S + \omega_o^2}$$

$$Q = 1 + R_1/R_2$$

$$f_{\text{notch}} = \frac{\omega_o}{2\pi} = \frac{I_T}{2\pi RC I_O} = \frac{1}{2\pi RC} (R_O/R_T) (V_T/V_O)$$

Fig. 6. Second-order state variable filter.

$$T = RC \left(\frac{R_T}{R_O} \right) \left(\frac{V_O}{V_T} \right) \quad (5)$$

$$\omega_o = \frac{1}{RC} \left(\frac{R_O}{R_T} \right) \left(\frac{V_T}{V_O} \right) \quad (6)$$

Notice that the product of the terms I_O and R dictates how large a differential signal can be applied across the input terminals. The maximum differential signal V_{dmax} that the integrator can take on is limited by

$$V_{dmax} = I_O R = \left(\frac{V_O}{R_O} \right) R \quad (7)$$

In the actual circuit, R_O is matched to R so that $V_{dmax} = V_O$. V_O must be greater than the largest peak voltage that the integrator is expected to handle.

Although this technique does eliminate the problems mentioned earlier, it possesses several drawbacks that might have adverse effects on several key filter parameters. In general, circuits that employ low transconductance input stages tend to suffer from high noise, large offset voltage, and poor PSRR. This integrator design is no exception to these problems. In the following sections, these problems and ways to alleviate them are discussed in more detail.

IV. FILTER DESIGN AND PRACTICAL DESIGN CONSIDERATIONS

A. Second-Order State Variable Filter

A second-order state variable filter can be made by using two of these integrators as shown in Fig. 6. This particular configuration was chosen because it provides three functions— notch, bandpass, and low-pass responses—despite its simplicity. By making the two integrator time constants identical, the notch frequency at V_1 , the center frequency of the bandpass response at V_2 and the bandwidth of the low-pass response at V_3 are all given by the inverse of integration time constant, or ω_o . This critical frequency ω_o was given by (6) and rewritten for convenience:

$$\omega_o = \frac{1}{RC} \left(\frac{R_o}{R_T} \right) \left(\frac{V_T}{V_o} \right). \quad (6)$$

Consequently, it can be said that the frequency response of the filter is made voltage tunable. By changing the ratio V_T/V_o , ω_o can be varied to a desired value externally.

B. Effects of the Nonideal Multiplier Circuit

It is well known that the multiplier circuit as shown in Fig. 3 suffers from problems associated with the finite ohmic resistances and current gain (β) of the transistors used.

The finite ohmic resistances of n-p-n transistors cause the output current (I_{out}) of the transconductance stage to be a nonlinear function of signal current (i_s) and consequently it will manifest itself as a distortion at the output of the filter. For a special purpose filter in which the ratio of two currents I_o and I_T is fixed, the problem can be easily overcome by scaling the emitter areas of the transistors in proportion to the drive currents [4]. However, for a general purpose design this ratio is arbitrarily set by the user anywhere within the tuning range. For this reason these ohmic resistances must be minimized. In the actual filter circuit this is done by making the emitter contacts large and interdigitizing them with the base contacts. Also the emitter areas of Q_5 - Q_8 are scaled in such a way that at the middle of the tuning range (i.e., $V_T/V_o \approx \frac{1}{2}$) current densities of these devices are identical giving zero distortion at that point.

The effect of finite β of the n-p-n and p-n-p transistors is to impair the otherwise linear relationship between ω_o and V_T/V_o . While the lower β of p-n-p transistors has a more adverse effect on the linear relationship, the β error of n-p-n transistors is more important because of its dependence on temperature. Therefore, its effect will be fully analyzed in the following discussion.

The exact expression for I_{out} involving finite β of n-p-n transistors β_N is

$$I_{out} = \frac{(1 - 2X) - \frac{2G_o}{1 + \beta_N}}{1 + \frac{1 + 2G_o}{\beta_N}} I_T \quad (8)$$

where

$$G_o = I_T/2I_o = \frac{1}{2} \left(\frac{R_o}{R_T} \right) \left(\frac{V_T}{V_o} \right) \quad (9)$$

$$X = \frac{1}{2} \left(1 + \frac{i_s}{I_o} \right)$$

and the p-n-p β is assumed infinite.

Thus the effective transconductance g_{meff} also becomes dependent upon β_N :

$$g_{meff} = \frac{dI_{out}}{dV_d} = \frac{I_T}{I_o R} \left(1 - \frac{1 + 2G_o}{\beta_N} \right) \quad (10)$$

$$= \underbrace{\frac{R_o}{R_T} \frac{1}{R} \left(\frac{V_T}{V_o} \right)}_{\text{Ideal}} \left(1 - \underbrace{\frac{1 + 2G_o}{\beta_N}}_{\text{Error}} \right) \quad (11)$$

TABLE I
ERROR IN g_{meff} VERSUS CHANGES IN G_o AT VARIOUS VALUES OF β

$G_o = I_T/2I_o$	Error (%)				
	$\beta=50$	100	200	400	800
0.001	-2.0	-1.0	-0.50	-0.25	-0.125
0.01	-2.0	-1.0	-0.51	-0.26	-0.128
0.1	-2.4	-1.2	-0.60	-0.30	-0.150
1.0	-6.0	-3.0	-1.50	-0.75	-0.375

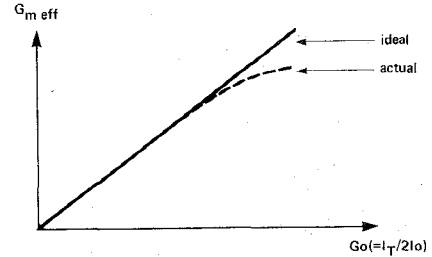


Fig. 7. Variation in g_{meff} versus G_o .

Since the error term in (10) or (11) is also dependent upon G_o ($\propto V_T/V_o$), g_{meff} will deviate from its ideal case by a different amount depending upon where the ratio V_T/V_o is set [see (9)]. Table I summarizes the amount of deviation for different values of β_N as G_o is changed by three orders of magnitude. At any value of β_N , g_{meff} will typically lag the ideal case progressively more as G_o is increased as shown in Fig. 7. For good tuning linearity between ω_o and (V_T/V_o) , higher β_N is required.

The lower β of p-n-p transistors as mentioned previously also causes nonlinear characteristics between ω_o and (V_T/V_o) . The effective transconductance is, assuming $\beta_N = \infty$, given by

$$g_{meff} = -\frac{I_T}{I_o} \left(1 - \frac{3}{\beta_P} \right) \quad (12)$$

$$= -\underbrace{\frac{R_o}{R_T} \frac{1}{R} \left(\frac{V_T}{V_o} \right)}_{\text{Ideal}} \underbrace{\left(1 - \frac{3}{\beta_P} \right)}_{\text{Error}}$$

Although the error term does not explicitly involve G_o , β_P is a function of collector current and as such the effective transconductance shows a strong dependence on G_o (i.e., V_T/V_o or I_T/I_o). Care must be taken to achieve high β_P as well as β_N . In reality, however, both β_N and β_P will be finite and will show dependence on current level, and a certain amount of nonlinearity can not be avoided. While this can be tolerated in most applications a rather strong dependence of β_N on temperature does affect another very important filter parameter and this will be discussed in the next section.

C. Drift of the Filter Response

One of the most important parameters of a stand-alone filter is the stability of its frequency response with temperature. It is far more important than the linearity between ω_o and V_T/V_o discussed in the previous section. Its first-order

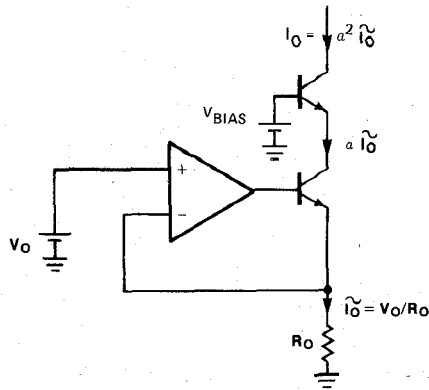


Fig. 8. V_O/I_O converter that compensates for linearity error caused by β variation.

dependence on temperature is clear from (6). Since the temperature coefficients (TC's) of on-chip resistors will track with each other, the temperature drift of ω_o is the inverse of the sum of the TC's of the resistor and the capacitor. The TC of the silicon chrome thin film resistor used to form the resistors is about +100 ppm/°C and that of silicon nitride capacitor is about +20 ppm/°C. Consequently, the filter drift can be expected to be about -120 ppm/°C due to R and C .

There are other second-order effects that contribute to the drift. One of the significant contributions could come from the temperature dependence of β_N as mentioned earlier. When this effect is taken into account, from (10) and (11),

$$\omega_o = \frac{g_{m\text{eff}}}{C} = \left(\frac{I_T}{I_o} \right) \left(\frac{1}{RC} \right) \left(1 - \frac{1 + 2G_o}{\beta_N} \right) \quad (13)$$

$$= \underbrace{\frac{R_o}{R_T RC} \left(\frac{V_T}{V_o} \right)}_{\text{Ideal}} \underbrace{\left(1 - \frac{1 + 2G_o}{\beta_N} \right)}_{\text{Error}} \quad (14)$$

A quick look at Table I reveals that the error varies with β_N . At any G_o the amount of deviation from ideal (which is in a direction to reduce ω_o) decreases as β_N increases. Since β_N increases with temperature, the actual ω_o approximates the ideal case more closely as temperature rises. The temperature drift due to β_N variation will be most pronounced when $G_o = 1$ as is obvious from (13) (error = $-3/\beta_N$). This problem can be easily alleviated by designing I_o to have a positive TC as shown in Fig. 8. In this case,

$$I_o = \left(\frac{V_o}{R_o} \right) \alpha^2. \quad (15)$$

where α is the common base current gain of n-p-n transistors. Substituting (15) in (13) yields

$$\omega_o = \frac{R_o}{R_T} \frac{1}{RC} \frac{V_T}{V_o} \frac{1}{\alpha^2} \left(1 - \frac{1 + 2G_o}{\beta_N} \right). \quad (16)$$

In terms of β_N , (16) simplifies to

$$\omega_o \approx \underbrace{\frac{R_o}{R_T} \frac{1}{RC} \frac{V_T}{V_o}}_{\text{Ideal}} \underbrace{\left(1 + \frac{1 - 2G_o}{\beta_N} \right)}_{\text{Error}}. \quad (17)$$

At $G_o = 0.001$, the error term $\approx 1/\beta_N$ and at $G_o = 1$, $-1/\beta_N$. Thus, it is clear that the temperature dependence is reduced to less than $\pm 1/\beta_N$ at any G_o . For a typical β_N of 200 and its drift of 0.5 percent/°C, one may expect this contribution to the filter to be 25 ppm/°C.

The drift of the offset voltages of the op amps used to do V_T/I_T and V_o/I_o conversions will also contribute to the drift of ω_o which is most pronounced at low values of V_T and V_o . A careful layout of these op amps is very important to minimize these effects.

To summarize, the drift of the filter is dominated by the TC's of R and C . Due to the good design and careful layout secondary effects are minimized to an insignificant level.

D. Dynamic Range of the Filter

In order to maximize the dynamic range of the filter the noise of the integrators must be minimized. Of the transistors used in the transconductance amplifier (see Fig. 5) Q_6 , Q_7 , Q_9 , Q_{10} are the main contributors of noise. This is due to the fact that the transconductance from input to the collectors of Q_2 and Q_3 ($\approx 1/R$) is further reduced to the outputs of the amplifier by the ratio equal to I_o/I_T . Thus it is obvious that the current noise in Q_6 , Q_7 , Q_9 , Q_{10} reflects to the input as a large voltage noise. The current noise density $\tilde{i}$ in each one of these devices is given by

$$\tilde{i} = \sqrt{2q(I_T/2)} = \sqrt{qI_T} \quad (18)$$

where q is the electronic charge.

So the total contribution from the four devices $\tilde{i}_T$ is

$$\tilde{i}_T = \sqrt{4} \times \sqrt{qI_T} = 2\sqrt{qI_T}. \quad (19)$$

The input referred voltage noise density $\tilde{V}_T$ is given by

$$\tilde{V}_T = \frac{\tilde{i}_T}{g_{m\text{eff}}} = \frac{2\sqrt{qI_T}}{\frac{1}{R} \frac{I_T}{I_o}} = 2RI_o\sqrt{q/I_T}. \quad (20)$$

From (7),

$$\tilde{V}_T = 2V_{d\text{max}}\sqrt{q/I_T}$$

or

$$\frac{V_{d\text{max}}}{\tilde{V}_T} = \frac{1}{2} \sqrt{I_T/q}. \quad (21)$$

Equation (21) is a measure of the dynamic range that can be achieved by the filter. It shows that the dynamic range will increase as a square root of I_T . Obviously, the increase in I_T must be accompanied by the proportional increase in the size of the integrating capacitor in order to maintain the same time constant of the integrator. In other words, the dynamic range will increase as a square root of the capacitor size. The high dielectric constant of silicon nitride is taken advantage of in forming a pair of large capacitors in the actual filter.

Another important factor in obtaining large dynamic range is to utilize the full rail-to-rail voltage available. That is, $V_{d\text{max}}$ should always be set as high as possible for a given set of power supply voltages. Since the increases in $V_{d\text{max}}$ ($=I_oR_o$) must be accompanied by the increases in I_T to maintain the same

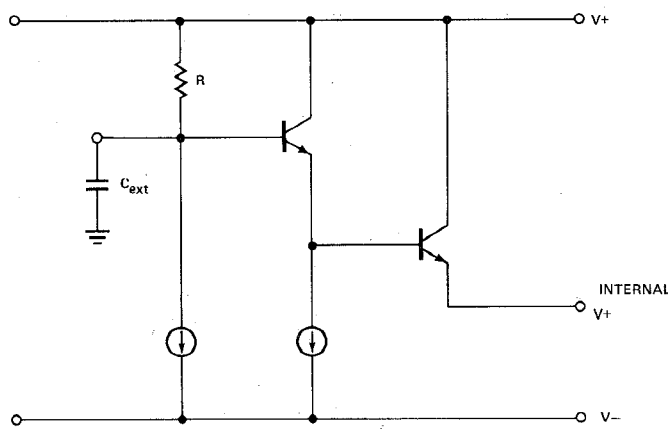


Fig. 9. Supply noise rejection circuit.

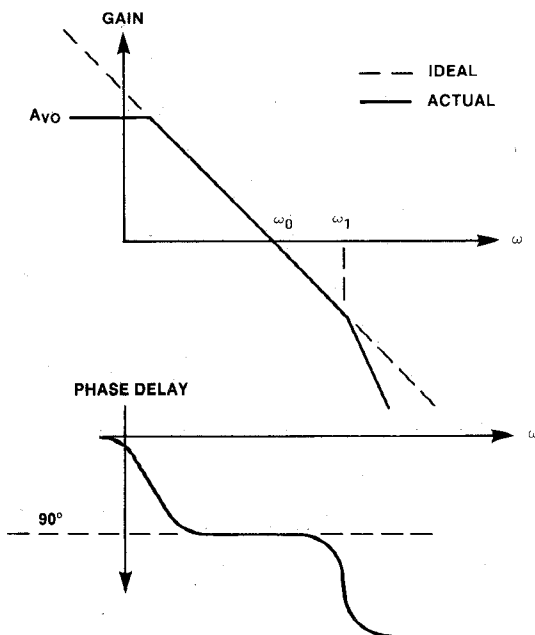


Fig. 10. Frequency response of integrator.

time constant, this results, according to (21), in an increase of dynamic range.

Noise on the supply lines also couples into the filter rather easily, again due to low transconductance of the input stage. A simple low-pass filter is employed to reject noise on the supply line as shown in Fig. 9. The noise on the positive supply line is attenuated by the RC time constant. The negative supply line is designed similarly.

E. Accuracy of Frequency Response

The accuracy of the filter response is limited by the non-idealities of the integrators used in the filter. An ideal integrator would have a constant 90° phase lag at all frequencies. The actual integrator would approximate this fairly well in the middle of the frequency range that it is designed to handle, but at the lower end of the frequency range this phase lag is less than 90° due to finite dc gain. At the higher end, the lag is greater than 90° due to nondominant poles that unavoidably exist (see Fig. 10). A simple calculation reveals that at the lower frequency edge, the notch depth is limited by the finite dc gain A_v to $2Q/A_v$. For $Q = 2$, $A_v > 400$ is required for

40 dB of notch depth. Both bandpass and low-pass responses tend to flatten at ω_o by roughly $2Q/A_v$. For these reasons bias currents must be carefully designed to achieve sufficient dc gain. As for the effects of the secondary poles, excessive phase delay that they introduce tends to reduce notch depth as well as to cause peaking in the bandpass and low-pass responses. In this case the calculations are somewhat more involved. If we simply model the integrator response $H(s)$ by

$$H(s) = \frac{\omega_o}{s} \frac{1}{1 + \left(\frac{s}{\omega_1}\right)},$$

where ω_o is the ideal unity gain frequency and ω_1 is the location of the parasitic pole, then it can be shown for the filter with $Q = 2$ that in order to achieve the notch depth > 40 dB and the peaking in the bandpass and low-pass responses < 0.1 dB we must have $\omega_1/\omega_o > 400$. It is very important, therefore, to optimize the frequency response of the integrator. In particular, the p-n-p transistor (Q_{12} in Fig. 5) which limits the frequency response must be operated at

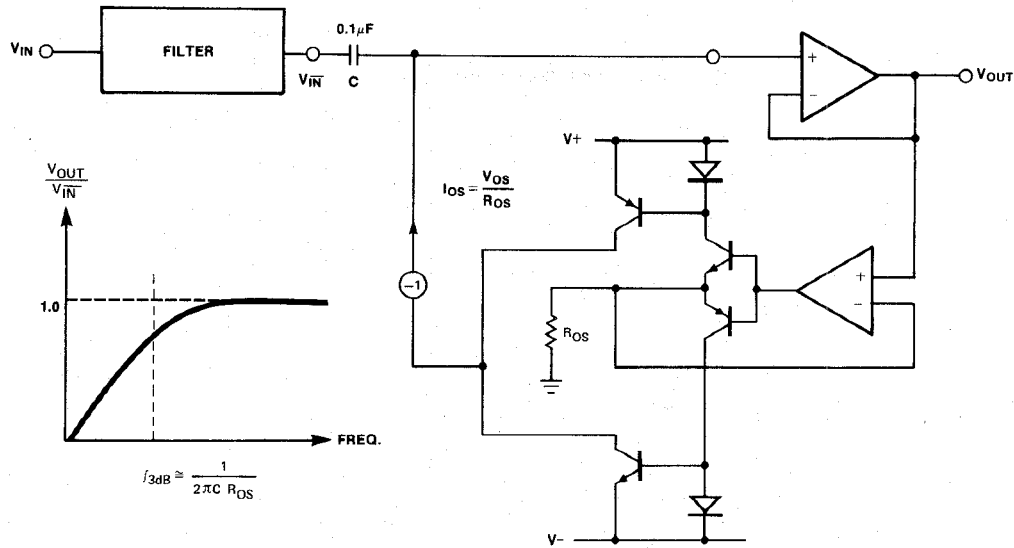


Fig. 11. Output offset correction circuit.

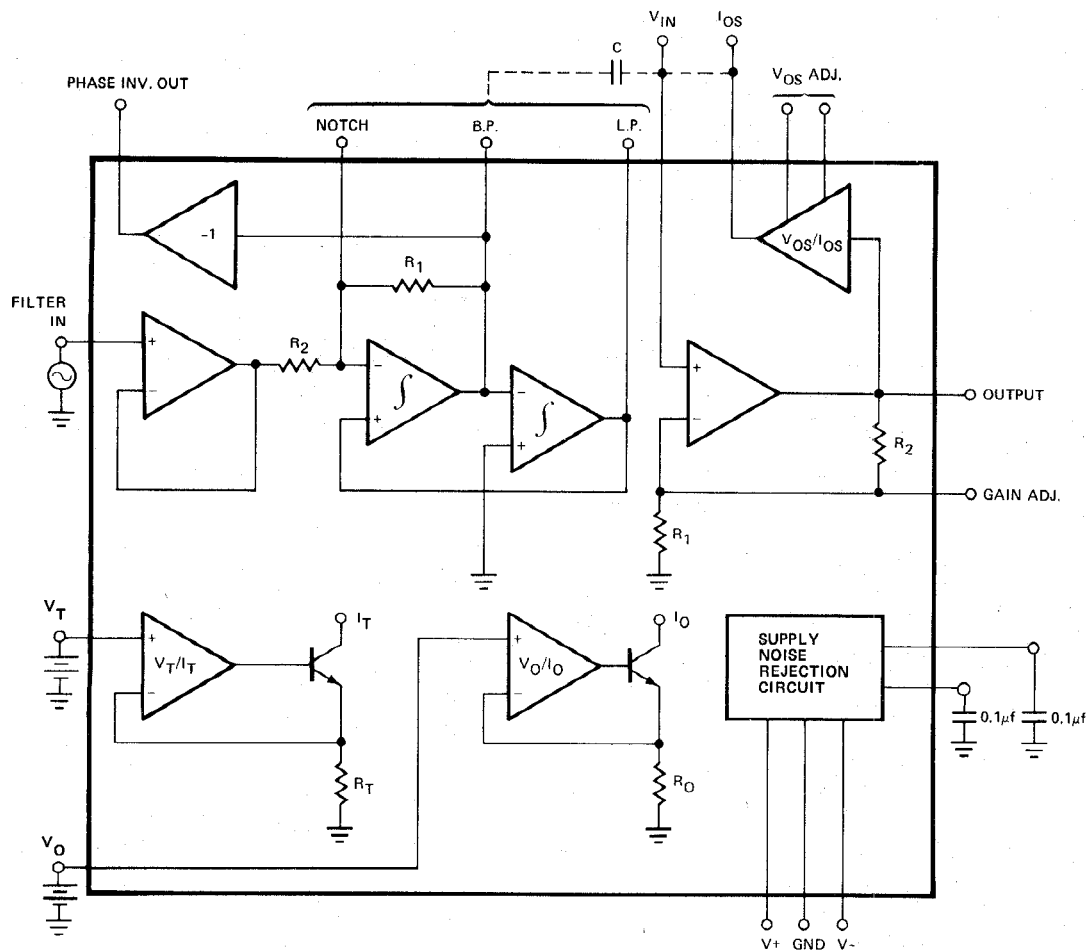


Fig. 12. Block diagram of the tunable filter.

the collector current where f_T is high. The p-n-p transistors Q_9 - Q_{10} in Fig. 5 have their base width reduced to the minimum for better frequency response.

The absolute accuracy of the filter response can be adjusted by trimming either R_O or R_T [see (6)]. Since R_O is matched to R to set V_{dmax} , R_O should be left alone. R_T therefore is the best candidate for trimming. R_T is nominally made large

enough to accommodate process variations in thin film resistors and capacitors and then Zener-trimmed [6] at wafer sort. An interesting possibility would be to make R_T external. By doing so, the filter response can be made user-programmable. Furthermore, if its TC is very small ($\ll 100$ ppm/ $^{\circ}$ C) the dominant contribution term to the filter drift can be mostly removed.

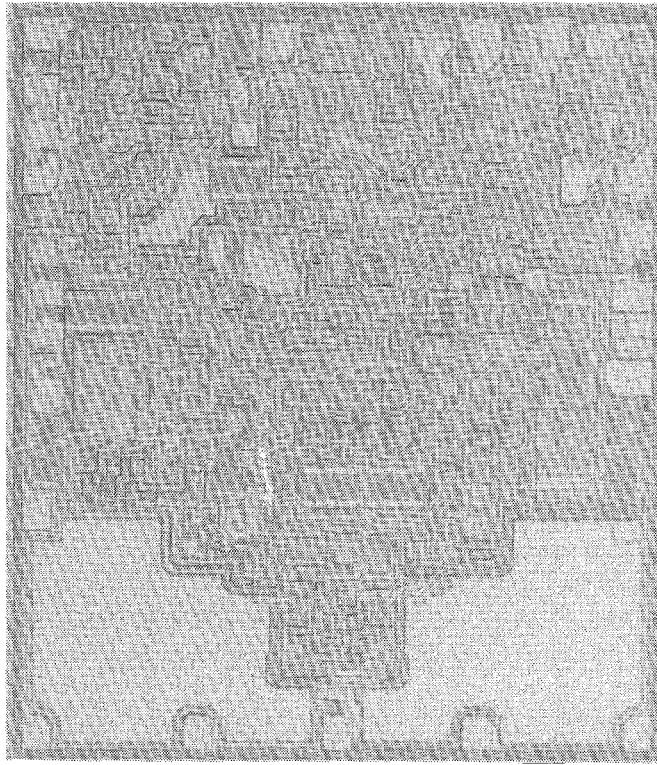


Fig. 13. Die photo of the tunable filter.

TABLE II
TYPICAL FILTER PERFORMANCE
@ $T = 25^{\circ}\text{C}$, $V_S = \pm 15\text{ V}$, $Q = 2$

PARAMETER	TYPICAL PERFORMANCE
POWER CONSUMPTION	200mW
ABSOLUTE ACCURACY	$\pm 3\%$
TUNING LINEARITY (IN ANY OCTAVE)	$\pm 1\%$
MAXIMUM INPUT SIGNAL @ $V_O = 10\text{V}$.	10Vp
NOISE (10Hz - 100kHz):	220 μV
@ $V_O = 10\text{V}$, $V_T = 2\text{V}$ NOTCH	
BANDPASS	150 μV
LOWPASS	150 μV
DISTORTION @ BANDPASS OUTPUT:	
@ $V_O = 5\text{V}$, $V_T = 1\text{V}$ $V_{IN} = 8\text{Vp}$	0.3%
$V_{IN} = 1\text{Vp}$	0.1%
NOTCH DEPTH:	
@ $V_O = 5\text{V}$, INPUT FREQUENCY = $\omega_O = 200\text{Hz}$ ($V_T \approx 50\text{mV}$)	50dB
= 2KHz ($V_T \approx 500\text{mV}$)	60dB
= 10KHz ($V_T \approx 2.5\text{V}$)	42dB
BANDPASS AND LOWPASS RESPONSE ACCURACY:	
@ $V_O = 5\text{V}$, INPUT FREQUENCY = $\omega_O = 200\text{Hz}$ ($V_T \approx 50\text{mV}$)	-0.05dB
= 2KHz ($V_T \approx 500\text{mV}$)	-0.02dB
= 10KHz ($V_T \approx 2.5\text{V}$)	+0.08dB
TEMPERATURE COEFFICIENT OF ω_O :	
@ $V_O = 5\text{V}$, $V_T = 1\text{V}$	-200ppm/ $^{\circ}\text{C}$
PSSR ⁺ , - at dc	40dB
PSSR ⁺ , - at 2KHz	50dB

F. Output Offset Voltage

The low transconductance input amplifier used in the design of integrators introduces high offset voltage. For example, 1 mV of offset voltage between Q_5 and Q_8 would reflect to the input as 200 mV of offset voltage at room temperature if RI_O is set to 10 V. To alleviate this problem, the filter output

is, through an external capacitor, ac-coupled to the output buffer (see Fig. 11). In order to provide a necessary dc bias to the buffer input, its dc offset voltage is converted to an offset correction current I_{os} by a voltage to current converter (V_{os}/I_{os} converter) and fed to the buffer input. At dc this feedback loop helps maintain low output voltage. For all relevant ac frequencies, I_{os} is shunted into the filter output by the cou-

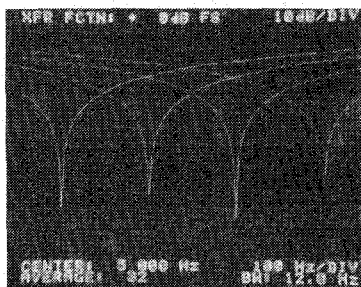


Fig. 14. Example of the filter being used as a notch filter as V_T/V_O is varied.

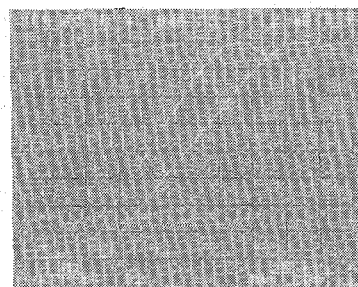


Fig. 15. Example of the filter being used as a bandpass filter V_T/V_O is varied.

pling capacitor C . The 3 dB point f_{3dB} of this ac coupling is given by

$$f_{3dB} = \frac{1}{2\pi R_{os} C}$$

where R_{os} is an on-chip resistor used to convert V_{os} to I_{os} and C is the external capacitor used. With $C = 0.1 \mu F$ it is around 10 Hz and in most applications this should not be a problem.

G. Complete Filter Design

Fig. 12 is a block diagram of the complete filter showing the interconnection of the various blocks so far discussed. A phase inverter is also included to enable the filter to be used as a voltage controlled oscillator. The frequency of oscillation is identical to the ω_o of the integrators.

V. EXPERIMENTAL RESULTS

A die photo of the complete filter is shown in Fig. 13. Six op amps and two integrators including about 1000 pF of capacitance are integrated in 11.4K mils² of silicon area. Table II summarizes the performance of the filter configured for $Q = 2$. The dynamic range of over 90 dB is obtained for the notch response, 94 dB for the low-pass and bandpass responses. The temperature drift of the filter is less than 200 ppm/°C, most of which is due to the thin film (silicon chrome) used. In the middle of the tuning range (20 Hz–20 kHz) notch depth >60 dB is achieved and the bandpass and low-pass responses are within 0.02 dB of theoretical value. At 200 Hz, both the bandpass and the low-pass responses deviate (flatten) from theoretical value by about 0.05 dB while at 10 kHz, they show roughly 0.08 dB peaking above theoretical value.

Figs. 14 and 15 are examples of the filter being used as a notch and bandpass filter, respectively, as the ratio ($=V_T/V_O$) is varied.

VI. CONCLUSION

It has been shown that it is possible to design complete integrated precision filters using only bipolar technology. The concepts discussed have been applied to a voltage tunable state

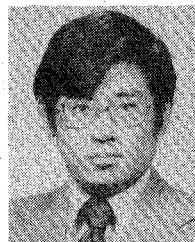
variable filter requiring no external precision components, but should also prove useful in many other filter designs.

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